ABDERRAHMAN EL FARCHOUNI

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SUMMARY

PhD candidate in Hydrology, isotopes, hydrochemistry and geophysics at University Mohammed VI Polytechnique (UM6P - IWRI) and Università degli Studi di Firenze (UNIFI - DAGRI) through a cotutelle program. My research focuses on groundwater recharge and water resources management, combining hydrological modeling, remote sensing, geophysics, hydrochemistry, isotopes, and machine learning to improve hydrogeological conceptual models in semi-arid regions. I am experienced in both fieldwork and data analysis, with the ability to design multidisciplinary research frameworks that support sustainable water governance.

EDUCATION

PhD — Oct 2023 - Present

- Research focus: Underground hydraulic aquifer systems of the piedmont of the Atlas of Marrakech : hydrogeophysical assessment and modeling

Master — 2021-2023

Faculty of Sciences, Kenitra

- Master Hydroinformatic and Hydrosystems Management

Bachelor in Applied Geosciences — 2017-2021

Faculty of Science and Technology, Tangier – Morocco

- Applied Geosciences

DEUST — 2017-2020

Faculty of Science and Technology, Tangier – Morocco

- Foundation degree (Bac+2) providing multidisciplinary scientific background.

Baccalaureate — 2020

- Baccalaureate in Mathematics A

Baccalaureate — 2017

- Life and Earth Sciences

RESEARCH EXPERIENCE

PhD Researcher – Hydrology, hydrochemistry, isotopes and Hydrogeophysics — Oct 2023 - Present

Mohammed VI Polytechnic University (UM6P), International Water Research Institute (IWRI)

- Conducting research on groundwater recharge and conceptual modeling of the Haouz–Mejjate aquifer system (Atlas of Marrakech).
- Integration of hydrochemistry, stable isotopes, geophysics, and remote sensing to assess recharge dynamics in semi-arid regions.
- Multidisciplinary framework combining field campaigns, data analysis, and hydrological modeling.
- Integrated Hydrological Modelling

Fieldwork and Data Collection:

- **Geophysical surveys:** Electrical Resistivity Tomography (ERT), Magnetic Resonance Sounding (MRS), seismic surveys, and EM38.
- **Hydrogeological monitoring:** Piezometric network measurement and data analysis.
- **Water sampling:** Collection and analysis of isotopic (δ^2H, $\delta^{18}O$) and hydrochemical samples.
- **Aquifer characterization:** Execution and interpretation of pumping tests.
- **Socio-economic surveys:** Design and implementation of farmer questionnaires on irrigation practices, groundwater recharge, and water management strategies.

PUBLICATIONS

Peer-Reviewed Journal Articles:

[1] El Farchouni, A., Hadri, A., Ait Brahim, Y., Hssaisoune, M., Bouchaou, L., Ouarani, M., Miftah, M., Bouimouass, H., Castelli, G., Bresci, E., Penna, D., Kchikach, A., & Fakir, Y.. "Groundwater recharge mechanisms in the Haouz-Mejjate aquifer (Morocco): Insights from environmental tracers and self-organizing maps." Groundwater for Sustainable Development, 101706. https://doi.org/10.1016/j.gsd.2026.101706, 2026. Data and code: https://doi.org/10.5281/zenodo.23419569

[2] El Farchouni, A., Hadri, A., Fakır, Y., Ouarani, M., Azaroual, M., & Kchikach, A.. "Mapping groundwater recharge potential zones in a semi-arid, anthropogenically modified mountainous basin." Scientific African, 30(October), e03025. https://doi.org/10.1016/j.sciaf.2025.e03025, 2025.

[3] El Mezouary, L., Elfarchouni, A., Hadri, A., Hakim, M., Fakır, Y., Bouchaou, L., & Chehbouni, A.. "Geostatistical modeling of subsurface heterogeneity in the Al-Haouz-Mejjate aquifer system, Morocco: A T-PROGS approach for enhanced hydrogeological characterization." Scientific African, 31, e03163. https://doi.org/10.1016/j.sciaf.2025.e03163, 2026.

[4] El Mezouary, L., Kharrou, M. H., Hadri, A., Elfarchouni, A., Fakir, Y., Bouchaou, L., & Chehbouni, A.. "Advanced hydrogeological modeling to reduce uncertainty in groundwater recharge estimation in semi-arid aquifers: Case of Al-Haouz-Mejjate aquifer in Morocco." Chemosphere, 392, 144743, 2025.

[5] El Mezouary, L., Hadri, A., Kharrou, M. H., Fakır, Y., Elfarchouni, A., Bouchaou, L., & Chehbouni, A.. "Contribution to advancing aquifer geometric mapping using machine learning and deep learning techniques: a case study of the AL Haouz-Mejjate aquifer, Marrakech, Morocco." Applied Water Science, 14(5).

Conference Proceedings

[6] El Mezouary, L., Hadri, A., Kharrou, M.H., Fakir, Y., Elfarchouni, A., Bouchaou, L., Chehbouni, A.. "Machine Learning and Deep Learning Guided Assessment of Groundwater Reservoir Hydrodynamic Parameters: A Case Study of The El Haouz Aquifer." 4th International GIRE3D Congress on Participatory and Integrated Management of Water Resources in Arid Zones, Laayoune, Morocco, Nov 2023.

[7] El Farchouni, A., et al.. "Mapping Groundwater Recharge Potential Zones Using Remote Sensing, GIS, and AHP Techniques: A Case Study of the Tensift Basin, Morocco." IAHR Africa Congress, Dec 9–12, 2024, 2024.

[8] El Farchouni, A., Hadri, A., Castelli, G., Fakir, Y., Bresci, E., Ouarani, M., & Kchikach, A.. "Integrating SWAT+ hydrological modelling and remote sensing analysis to estimate surface water balance and groundwater recharge in the High Atlas of Marrakech and the Haouz plain (Morocco)." EGU General Assembly 2026, Vienna, Austria, May 3–8, 2026 (oral presentation, session HS8.2.13, abstract EGU26-21149).

[9] El Farchouni, A., et al.. "Integrating SWAT+ hydrological modelling and remote sensing to estimate surface water balance and groundwater recharge in the High Atlas of Marrakech and the Haouz plain." 5th International Conference on African Rivers (ICAR 2026), Rabat, Morocco, Jul 6–11, 2026 (oral communication).

[10] El Farchouni, A., Hadri, A., Fakir, Y., Ait Brahim, Y., Hssaisoune, M., Bouchaou, L., Ouarani, M., Bouimouass, H., Castelli, G., Bresci, E., Penna, D., & Kchikach, A.. "Hydrochemical and Isotopic Assessment of Groundwater Recharge Mechanisms in Semi-Arid Alluvial Aquifers: Integrating Environmental Tracers with Conceptual Modeling." IAH 2026 World Groundwater Congress (53rd Congress of the International Association of Hydrogeologists), Budapest, Hungary, Sep 14–18, 2026 (poster presentation, supported by a UNESCO Travel Grant administered by IAH).

TECHNICAL SKILLS

Remote Sensing & GIS:

- **Google Earth Engine (GEE)** for large-scale geospatial analysis (land use/land cover, NDVI/NDWI, evapotranspiration modeling, groundwater recharge mapping).
- **ArcGIS, QGIS** for spatial analysis, interpolation, and cartography.
- Satellite data processing (Sentinel-2, Landsat, CHIRPS, WaPOR, TerraClimate, SMAP, GPM etc).

Hydrological & Hydrogeological Modeling:

- **SWAT/SWAT+ and MODFLOW** for integrated surface–groundwater modeling
- **GMS:** Groundwater Flow modeling

Geophysics & Field Methods:

- Electrical Resistivity Tomography (ERT), Magnetic Resonance Sounding (MRS), seismic surveys, EM38.
- Pumping tests, piezometric monitoring, isotopes ($\delta^{2}H$, $\delta^{18}O$) and hydrochemistry sampling.
- Geophysics software: Prodiviner, Res2DInv

Programming & Data Science:

- **Python** (pandas, scikit-learn, TensorFlow, Xarray, matplotlib etc).
- **Machine learning & deep learning** (KNN, Random Forest, LSTM, GAIN, PINN)
- **R for statistical and hydrogeochemical analysis** (hydrogeochemistry diagrams, PCA, clustering).
- **SQL & SQLite Studio** for database management.

Other Tools:

- **Adobe Illustrator** for scientific figure editing and publication-quality graphics.
- **Microsoft Office** Suite (Excel, Word, PowerPoint).
- **LaTeX** for academic writing and reports.

CONFERENCES WORKSHOPS & TRAINING

- **IAH 2026 World Groundwater Congress (53rd Congress of the International Association of Hydrogeologists), Budapest, Hungary:**
  - Poster presentation: Hydrochemical and Isotopic Assessment of Groundwater Recharge Mechanisms in Semi-Arid Alluvial Aquifers: Integrating Environmental Tracers with Conceptual Modeling (Sep 14–18, 2026).
  - Participation supported by a UNESCO Travel Grant, administered by IAH on behalf of UNESCO.
- **15th ESA Advanced Training Course on Land Remote Sensing: The Water Cycle, University of Perugia, Italy:**
  - European Space Agency advanced training on Earth Observation for the water cycle, using data from Copernicus Sentinels and other satellite missions (Sep 7–11, 2026).
- **5th International Conference on African Rivers (ICAR 2026), Rabat, Morocco:**
  - Oral communication: Integrating SWAT+ hydrological modelling and remote sensing to estimate surface water balance and groundwater recharge in the High Atlas of Marrakech and the Haouz plain (Jul 6–11, 2026).
  - Post-conference training: Machine Learning for Hydrological Impacts Mapping (Jul 10–11, 2026).
  - Post-conference training: Deep Learning for Hydrological Time Series Prediction (Jul 10–11, 2026).
- **EGU General Assembly 2026, Vienna, Austria:**

- Oral presentation: Integrating SWAT+ hydrological modelling and remote sensing analysis to estimate surface water balance and groundwater recharge in the High Atlas of Marrakech and the Haouz plain (Morocco), session HS8.2.13 (May 3–8, 2026).

- **IAHR Africa Congress:**
  - Mapping Groundwater Recharge Potential Zones Using Remote Sensing, GIS, and AHP Techniques: A Case Study of the Tensift Basin, Morocco (Dec 9–12, 2024).
- **Discover Scopus AI (May 15, 2024).**
- **Workshop on Isotopic Tracers in Hydrology and Earth Sciences** (May 28–29, 2024).
- **Karst and Cave Sustainable Management Summer Course** (Jul 29 – Aug 9, 2024).
- **Enhancing Water Productivity with WaPOR: Using WaPLUGIN in QGIS** (Dec 12, 2024).
- **Mathematical Models and Artificial Intelligence for Solving Groundwater Flow Problems** (Jan 6–7, 2025).
- **Maghrebian Colloquium of Applied Geophysics (MCGA-9) – Active participation in the 9th edition, Faculty of Sciences, Ibn Tofail University, Kenitra, Morocco** (Apr 23–25, 2024).
- **SYMPLE: Introduction to Applied Groundwater Flow Modelling with GW Vistas – online training** (Mar 17–19 & Apr 7–10, 2025).
- **SYMPLE: Understand MODFLOW** (May 6–27, 2025).

LANGUAGES

- **Arabic (Native)**
- **French (Fluent)**
- **English (Fluent)**